Bhanu Pratap Singh

Full-Stack Engineer · Founding Engineer · Distributed Systems & Cloud

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SUMMARY

Full-stack engineer who has taken EdTech and SaaS platforms from MVP to production at scale. As Tech Lead for Competishun (100K+ users, 50K+ peak concurrent test-takers), cut cloud cost by **40%**, backend CPU from **99% to 25%** and video processing from **6 hours to under 30 minutes**, and led a **14-microservice GCP → AWS migration**. Now founding engineer on Synapses, a white-label enterprise LMS. Works directly with founders, clients and domain experts to turn ambiguous requirements into shipped systems: Node.js, TypeScript, PostgreSQL, Redis, AWS.

EXPERIENCE

Newral Softwares Inc.

Dec 2024 – Present

Founding Engineer, Synapses · white-label enterprise LMS SaaS

- Building and maintaining the platform's microservices with **Next.js, TypeScript, PostgreSQL, Redis and BullMQ**; multi-tenant design lets each enterprise run the LMS under its own brand.
- Architected **multi-tenant payments** with Cashfree: per-tenant isolation, tokenized card vaulting (PCI-aware) and subscription billing.
- Run large-scale data operations across SQL and NoSQL databases and own production incident resolution.

Tech Lead, Competishun · EdTech platform, 100K+ users, 25K DAU · led team of 6 (4 engineers, 2 interns)

- Cut **cloud cost by 40%** and shrank compute from **32 to 4 cores** by optimizing SQL queries and consolidating cross-service network calls onto a single network path.
- Reduced **backend CPU from 99% to 25%** through query optimization, Redis caching and workload restructuring, keeping weekly test series stable at **50K+ concurrent students**.
- Led migration of **14 microservices from GCP to AWS** (via ACE Cloud), improving scalability, reliability and cost efficiency.
- Rebuilt the **FFmpeg video transcoding pipeline** (S3 presigned ingest, adaptive-bitrate HLS, DRM playback), cutting processing time from **6 hours to under 30 minutes**.
- Designed and built **4 of the platform's 14 microservices**; fixed critical production issues across services, cutting **app crashes by 90%**. Owned code review and release quality for the team.

Full-Stack Engineer, Client Delivery · startups, VC-backed and government-funded

- **Sole engineer on AssumeChat**: shipped MVP and V1 used in VC demos and the funding round.
- Delivered the **Agrevon** MVP (Startup India Seed Fund) and the **Rapid Stroke** backend, working directly with doctors and medical students on requirements.
- Set up Prometheus/Grafana monitoring and GitHub Actions CI/CD for **Fizz Social**; built **Emotions AI** on React Native and Next.js.

Eventory · Full-Stack Engineering Intern

Aug 2024 – Oct 2024

- Built a two-sided event marketplace with vendor SaaS subscriptions, customer discovery and booking; integrated **Razorpay and Stripe** payment flows.

PROJECTS

Vibe Social (vibesocial.bpss.tech): video and post sharing platform with subscriptions, playlists, analytics and an in-house FFmpeg transcoder that writes multiple resolutions to AWS S3. *React, Node.js, Express, MongoDB, Redux, AWS S3*

Linky (linky.bpss.tech): SaaS URL shortener with JWT auth, multi-target redirects and click analytics (IP, device, OS). *Next.js, Express, TypeScript, PostgreSQL, Prisma*

Unfiltered Byte (unfiltered-byte.bpss.tech): community and wellness mobile app for developers. *React Native, Expo, Node.js, MongoDB, Appwrite*

Engineering writing (bpss.in/blog): *Architecting a Resilient Video Transcoding Pipeline*; *Architecting Multi-Tenant Payments*.

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, SQL
Backend	Node.js, Express, Microservices, REST APIs, BullMQ, Redis, JWT / NextAuth
Databases	PostgreSQL, MongoDB, Prisma, Mongoose, query optimization & indexing
Cloud & DevOps	AWS, GCP, Docker, NGINX, GitHub Actions, Prometheus, Grafana, Vercel
Frontend & Mobile	React, Next.js, React Native (Expo), Redux, Zustand, Tailwind CSS, shadcn/ui
Media & Payments	FFmpeg, HLS / adaptive bitrate, DRM, Shaka Player, Cashfree, Razorpay, Stripe

EDUCATION

Bachelor of Commerce (B.Com), Maharshi Dayanand Saraswati University (MDSU), Ajmer — May 2026

Full-Stack Web Development (MERN, DSA, System Design), Masai School — Sep 2023 – Jul 2025